

H2 Water Technologies, Ltd. v. Echo Water

The public complaint analysis describes a composition patent for producing hydrogen-rich water from a powder or tablet. The alleged claim-1 composition uses magnesium metal, acid, and a non-cellulose polysaccharide and requires at least 3 mM hydrogen after contact with 500 mL of water at atmospheric pressure and room temperature. The preliminary issue map therefore centers on formulation chemistry, powder and particle characterization, acid-metal reaction kinetics, dissolved-hydrogen measurement, and carbohydrate classification.

Independent preliminary research based on public information.

FILED	FORUM	DOCKET	MATTER
2026-08-24	U.S. District Court for the District of Delaware	1:26-cv-01078	Patent Infringement

Prepared 2026-08-27. No attorney or expert contacted. Availability, interest, compensation, and conflicts not checked.

Public filing, likely recipient, and technical frame

CAPTION H2 Water Technologies, Ltd. v. Echo Water LLC

FORUM / DOCKET U.S. District Court for the District of Delaware / 1:26-cv-01078

FILED / TYPE 2026-08-24 / patent infringement

PUBLIC DOCKET [Open docket](#)

VERIFIED PUBLIC RECORD

Public docket reporting identifies H2 Water Technologies, Ltd. v. Echo Water LLC, Civil Action No. 1:26-cv-01078, filed August 24, 2026 in the District of Delaware—not the Northern District of Illinois. The docket identifies a jury-demand patent-infringement complaint, U.S. Patent No. 12,588,693, and plaintiff counsel Richard C. Weinblatt and Stamatios Stamoulis. The public complaint analysis reports allegations concerning Echo powdered drink mixes that generate dissolved molecular hydrogen. Allegations are not findings.

LIKELY RESEARCH PURCHASER

Stamatios Stamoulis - Attorney, Stamoulis & Weinblatt LLC. Verified docket fact and reasonable outreach prioritization: the public docket identifies Stamoulis as plaintiff counsel in this newly filed patent case. His firm biography describes more than 20 years in intellectual-property and complex-commercial matters and patent-infringement litigation. That makes him a plausible attorney account for a preliminary expert-search packet; it does not establish purchasing authority.

stamoulis@swdelaw.com - publicly printed in the [linked professional source](#); not inferred.

TECHNICAL FRAME

The public complaint analysis describes a composition patent for producing hydrogen-rich water from a powder or tablet. The alleged claim-1 composition uses magnesium metal, acid, and a non-cellulose polysaccharide and requires at least 3 mM hydrogen after contact with 500 mL of water at atmospheric pressure and room temperature. The preliminary issue map therefore centers on formulation chemistry, powder and particle characterization, acid-metal reaction kinetics, dissolved-hydrogen measurement, and carbohydrate classification.

Secondary-source limitation: The docket establishes the filing and parties. The technical detail is drawn from a public secondary analysis of the complaint and must be checked against the filed complaint, patent, prosecution history, product samples, validated test protocols, and later discovery before any merits or infringement opinion. [Open public synopsis](#)

REPORTED ASSERTED PATENTS

[U.S. Patent No. 12,588,693](#)

Questions likely to require expert input

This issue map is a research plan, not a legal conclusion. The filed complaint, patent exhibits, intrinsic record, and discovery must control.

01 Powder composition and ingredient identity

What are the actual ingredients, amounts, particle-size distributions, and manufacturing attributes of the accused powder, including the amount and mesh size of magnesium and the identities and amounts of acids and fiber components?

Potential evidence: Retained product samples, labels, certificates of analysis, bills of materials, supplier specifications, microscopy and sieving results, elemental analysis, batch records, and validated laboratory testing.

02 Magnesium-acid reaction kinetics

Under the patent's stated water volume, pressure, and temperature conditions, how do magnesium surface area, particle-size distribution, acid type, acid concentration, mixing, and pH affect the rate and amount of hydrogen generation?

Potential evidence: Reaction protocols, time-resolved gas measurements, pH and temperature measurements, particle characterization, reagent-purity records, replicates, uncertainty analysis, and laboratory notebooks.

03 Dissolved-hydrogen measurement

Does the accused product produce at least 3 mM dissolved molecular hydrogen after contact with 500 mL water at atmospheric pressure and room temperature, and what method validly distinguishes dissolved gas from headspace hydrogen or measurement artifacts?

Potential evidence: Pre-registered test protocol, sample handling records, dissolved-gas meters or gas chromatography methods, calibration data, temperature and pressure controls, headspace measurements, replicates, and error analysis.

04 Polysaccharide classification

Do fructooligosaccharides, xylooligosaccharides, chicory-root inulin, or other accused fiber ingredients meet the patent's claimed "a polysaccharide" limitation while not being cellulose or a cellulose derivative?

Potential evidence: Ingredient specifications, carbohydrate-composition testing, molecular-weight and degree-of-polymerization data, accepted carbohydrate nomenclature, intrinsic patent evidence, prosecution history, and expert testimony.

Questions likely to require expert input

05 Functional role of carbohydrate ingredients

What effect, if any, do the identified fiber or carbohydrate ingredients have on hydrogen generation, retention, bubble formation, mixing, pH, viscosity, powder dispersion, or product stability under the claimed conditions?

Potential evidence: Controlled formulation experiments, rheology and viscosity measurements, dissolution and dispersion testing, dissolved-gas time curves, comparative formulations, and product-development records.

06 Claim-condition reproducibility

Which experimental details materially affect results under the stated 500 mL, atmospheric-pressure, and room-temperature condition—including water chemistry, container geometry, agitation, sampling time, sealing, and instrument calibration—and were they controlled?

Potential evidence: Complete laboratory protocols, contemporaneous observation sheets, instrument logs, calibration records, photographs or video, blind replicates, sample chain of custody, and independent replication.

07 Product instructions and use

What conditions do Echo's labels and consumer directions specify, and do typical or instructed uses correspond to the asserted claim conditions?

Potential evidence: Product packaging, web archives, user instructions, retailer listings, marketing materials, customer-service scripts, design records, and testing conducted under instructed conditions.

08 Technical baseline and alternatives

What was known at the relevant time about on-demand hydrogen-water powders, magnesium-acid hydrogen generation, dietary-fiber formulations, and methods for producing or measuring dissolved hydrogen, and what technically feasible alternatives existed?

Potential evidence: Patent literature, peer-reviewed chemical and food-science papers, product literature, standards, prior-art formulations, laboratory reports, and a supported POSITA definition.

Candidates 1-2 for further diligence

Ranked for preliminary research priority based on subject-matter fit, relevant public work, testimony, and visible diligence burden. This is not a retention recommendation.

01 Lisa J. Mauer, PhD

Professor of Food Science, Purdue University

TECHNICAL FIT

Direct fit for powder formulation, water-solid interactions, hygroscopic behavior, dissolution, stability, and analytical testing of food and dietary-supplement ingredients.

RELEVANT WORK

Purdue identifies Mauer's work as food materials science, water-solid interactions, shelf-life, and infrared spectroscopy for non-destructive food analysis. A Purdue profile also describes work on ingredient architecture, formulation, processing, and stability of water-soluble ingredients.

PUBLIC LITIGATION EXPERIENCE

No clear public testifying role was located in this preliminary bounded search; this is not a representation that none exists.

POINTS FOR FURTHER DILIGENCE

The cited work is not specifically about the asserted patent or hydrogen-water products. Prior opinions, industry funding, consulting, and any party relationships require separate diligence.

[Source 1](#) | [Source 2](#)

[Draft email](#) | mauer@purdue.edu | [public email source](#)

02 Bruce R. Hamaker, PhD

Distinguished Professor of Food Science and Roy L. Whistler Chair, Purdue University

TECHNICAL FIT

Close fit for the claimed non-cellulose-polysaccharide limitation and the structure-function behavior of dietary fibers in a food composition.

RELEVANT WORK

Purdue identifies Hamaker as Director of the Whistler Center for Carbohydrate Research and describes expertise in carbohydrates, dietary fiber, and polysaccharide and protein structure-function relationships.

PUBLIC LITIGATION EXPERIENCE

No clear public testifying role was located in this preliminary bounded search; this is not a representation that none exists.

POINTS FOR FURTHER DILIGENCE

The record establishes carbohydrate expertise, not a view on the claim construction or the accused ingredient list. Scientific classification should remain tied to the actual formula and intrinsic evidence.

[Source 1](#) | [Source 2](#)

[Draft email](#) | hamakerb@purdue.edu | [public email source](#)

Candidates 3-4 for further diligence

03 John N. Coupland, PhD

Professor of Food Science, Pennsylvania State University

TECHNICAL FIT

Useful for physical chemistry of aqueous food systems, particle dispersion, stability, and formulation behavior when a powder is added to water.

RELEVANT WORK

Penn State identifies Coupland as a Food Science professor whose research includes physical properties of foods and emulsion science, with work on particles and delivery of compounds into aqueous foods.

PUBLIC LITIGATION EXPERIENCE

No clear public testifying role was located in this preliminary bounded search; this is not a representation that none exists.

POINTS FOR FURTHER DILIGENCE

The profile emphasizes physical properties of food, particularly emulsions, rather than magnesium-acid hydrogen generation. Pairing with a reaction-kinetics specialist may be appropriate.

[Source 1](#) | [Source 2](#)

[Draft email](#) | coupland@psu.edu | [public email source](#)

04 Job (Johan B.) Ubbink, PhD

Professor and Head, Food Science and Nutrition Department, University of Minnesota

TECHNICAL FIT

Strong fit for food-powder technology, phase behavior of carbohydrates, encapsulation, water uptake, chemical stability, and processing of instant beverage and nutritional powders.

RELEVANT WORK

University of Minnesota materials identify Ubbink's research in high-solids foods, food powders, encapsulation, starches, carbohydrates, and water uptake; his laboratory describes work on instant beverages and nutritional powder products.

PUBLIC LITIGATION EXPERIENCE

No clear public testifying role was located in this preliminary bounded search; this is not a representation that none exists.

POINTS FOR FURTHER DILIGENCE

Food-powder specialization does not by itself establish specific competence in dissolved-hydrogen instrumentation or magnesium-acid reaction kinetics.

[Source 1](#) | [Source 2](#) | [Source 3](#)

[Draft email](#) | jubink@umn.edu | [public email source](#)

Candidates 5-6 for further diligence

05 Lili He, PhD

Professor and Department Head, Food Science, University of Massachusetts Amherst

TECHNICAL FIT

Strong fit for analytical methods and food chemistry, particularly evaluating composition and chemistry of a commercial powdered food or nutraceutical product.

RELEVANT WORK

UMass identifies He as a professor and department head whose research develops analytical techniques for food-science issues, including surface-enhanced Raman spectroscopy and food chemistry applications.

PUBLIC LITIGATION EXPERIENCE

No clear public testifying role was located in this preliminary bounded search; this is not a representation that none exists.

POINTS FOR FURTHER DILIGENCE

The cited profile does not establish specialized dissolved-hydrogen assay experience. Method selection and validation would require matter-specific review.

[Source 1](#)

[Draft email](#) | lilihe@foodsci.umass.edu | [public email source](#)

06 Su Ha, PhD

Professor and George Austin Endowed Director, Washington State University

TECHNICAL FIT

Useful for reaction engineering, hydrogen generation, catalyst and reactor conditions, and quantitative analysis of gas-production mechanisms.

RELEVANT WORK

Washington State describes Ha's research as including production of hydrogen gas through thermochemical and electrochemical reforming, energy conversion, and fuel cells, with more than 100 publications in related energy research.

PUBLIC LITIGATION EXPERIENCE

No clear public testifying role was located in this preliminary bounded search; this is not a representation that none exists.

POINTS FOR FURTHER DILIGENCE

His public work concerns energy and fuel systems rather than ingestible powder formulations, so food-composition and analytical-method issues may require another expert.

[Source 1](#)

[Draft email](#) | suha@wsu.edu | [public email source](#)

Candidates 7-8 for further diligence

07 Anthony G. Dixon, PhD

Professor of Chemical Engineering, Worcester Polytechnic Institute

TECHNICAL FIT

Useful for reaction kinetics, gas generation, transport, particle-bed behavior, reactor modeling, and assessing how particle properties and reaction conditions affect measured hydrogen output.

RELEVANT WORK

WPI identifies Dixon's research in chemical reaction engineering and hydrogen production, including transport and reaction in particle beds, reactor modeling, and experimental and simulation studies of renewable hydrogen production.

PUBLIC LITIGATION EXPERIENCE

No clear public testifying role was located in this preliminary bounded search; this is not a representation that none exists.

POINTS FOR FURTHER DILIGENCE

His hydrogen-production research is directed to energy reactors rather than consumer beverage powders. The fit is strongest on kinetics and mass transfer, not food-label or carbohydrate classification.

[Source 1](#)

[Draft email](#) | agdixon@wpi.edu | [public email source](#)

08 Ravindra Datta, PhD

Adjunct Teaching Professor of Chemical Engineering, Worcester Polytechnic Institute

TECHNICAL FIT

Useful for chemical kinetics, catalysis, reaction networks, hydrogen and fuel-cell chemistry, and interpretation of quantitative gas-generation tests.

RELEVANT WORK

WPI describes Datta's three decades of work applying kinetics, catalysis, and reaction engineering to energy challenges, including hydrogen and fuel-cell research and published thermodynamic analysis of hydrogen production.

PUBLIC LITIGATION EXPERIENCE

No clear public testifying role was located in this preliminary bounded search; this is not a representation that none exists.

POINTS FOR FURTHER DILIGENCE

The current listed appointment is adjunct teaching professor, and his public work is energy-focused. Current role, prior engagements, and suitability for food-product testing need confirmation before any contact.

[Source 1](#)

[Draft email](#) | rdatta@wpi.edu | [public email source](#)

Candidates 9-10 for further diligence

09 Marcetta Y. Darensbourg, PhD

Distinguished Professor and Davidson Chair in Science, Texas A&M University

TECHNICAL FIT

Useful for molecular hydrogen chemistry, hydrogen evolution, and the chemical principles bearing on generation and detection of hydrogen in a reaction mixture.

RELEVANT WORK

Texas A&M identifies Darensbourg's research in bioinspired and inorganic chemistry and in catalysts for hydrogen production, with work focused on developing active hydrogen-producing catalysts.

PUBLIC LITIGATION EXPERIENCE

No clear public testifying role was located in this preliminary bounded search; this is not a representation that none exists.

POINTS FOR FURTHER DILIGENCE

The work is focused on molecular and catalytic chemistry, not consumer food formulation. Her fit is narrowest for the hydrogen-generation science, not product analytics or carbohydrate classification.

[Source 1](#)

[Draft email](#) | marcetta@chem.tamu.edu | [public email source](#)

10 Qing Zhao, PhD

Assistant Professor of Chemical Engineering, Northeastern University

TECHNICAL FIT

Useful for computational and molecular analysis of catalytic mechanisms and hydrogen-production chemistry, including how composition and material properties can influence reaction pathways.

RELEVANT WORK

Northeastern identifies Zhao's work in electronic-structure modeling and heterogeneous catalysis for sustainable energy, including hydrogen production and catalytic reaction mechanisms.

PUBLIC LITIGATION EXPERIENCE

No clear public testifying role was located in this preliminary bounded search; this is not a representation that none exists.

POINTS FOR FURTHER DILIGENCE

The cited research is computational and catalyst-oriented rather than beverage formulation or commercial-product testing, making this a secondary technical candidate rather than a first-line food-formulation expert.

[Source 1](#)

[Draft email](#) | q.zhao@northeastern.edu | [public email source](#)

Preliminary candidates; no conflicts or availability determination

- Run party, affiliate, counsel, inventor, funder, and technology conflicts.
- Confirm testimony history, exclusions, compensation, publications, patents, and deposition performance.
- Investigate licensing, grants, consulting, commercial interests, and prior technical positions.
- Confirm role fit, availability, and interest only after counsel authorizes outreach.

ONGOING MATTER-SPECIFIC RESEARCH

Flint Witness Research works in the post-filing window, turning pleadings and public technical records into issue maps, researched candidate slates, and diligence starting points as the case develops.

[Schedule a 15-minute conversation](#)

CORE SOURCES

Public docket: <https://dockets.justia.com/docket/delaware/dedce/1%3A2026cv01078/94370>

Public professional email source: <https://swdelaw.com/team/stamatios-stamoulis/>

Public technical context source: <https://ai-lab.exparte.com/case/dct/ded/1%3A26-cv-01078/doc/analysis/1>

LIMITATIONS

The docket establishes the filing and parties. The technical detail is drawn from a public secondary analysis of the complaint and must be checked against the filed complaint, patent, prosecution history, product samples, validated test protocols, and later discovery before any merits or infringement opinion.

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